



CLASSROOM CONTACT PROGRAMME
(Academic Session : 2019 - 2020)

NEET(UG)
MAJOR
17-02-2020

PRE-MEDICAL : ENTHUSIAST COURSE (ALL PHASE)

This Booklet contains 20 pages.

Do not open this Test Booklet until you are asked to do so.

Read carefully the Instructions on the Back Cover of this Test Booklet.

Important Instructions :

1. On the Answer Sheet and fill in the particulars on Side-1 and Side-2 carefully with blue/black ball point pen only.
2. The test is of 3 hours duration and this Test Booklet contains 180 questions. Each question carries 4 marks. For each correct response, the candidate will get 4 marks. For each incorrect response, one mark will be deducted from the total scores. The maximum marks are 720.
3. Use Blue/Black Ball Point Pen only for writing particulars on this page/markings responses.
4. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
5. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator before leaving the Room/Hall. The candidates are allowed to take away this Test Booklet with them.
6. The candidates should ensure that the Answer Sheet is not folded. Do not make any stray marks on the Answer Sheet. Do not write your Roll No. anywhere else except in the specified space in the Test Booklet/Answer Sheet.
7. Use of white fluid for correction is *not* permissible on the Answer Sheet.

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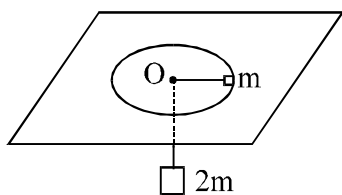
Candidate's Signature : _____ Invigilator's Signature : _____

Your Target is to secure Good Rank in Pre-Medical 2020

1. The resultant of $\vec{M}$ and $\vec{N}$ is perpendicular to $\vec{M}$.
The what is the angle between $\vec{M}$ and $\vec{N}$?

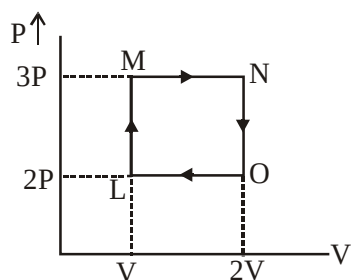
- (1) $\cos^{-1}\left(\frac{M}{N}\right)$ (2) $\cos^{-1}\left(-\frac{M}{N}\right)$
(3) $\sin^{-1}\left(\frac{M}{N}\right)$ (4) $\sin^{-1}\left(-\frac{M}{N}\right)$

2. A mass m rotating freely in a horizontal circle of radius 1 m on a frictionless smooth table supports a stationary mass $2m$, attached to the other end of the string passing through smooth hole O in table, hanging vertically. Find the angular velocity of rotation.



- (1) $\sqrt{5g}$ (2) $\sqrt{3g}$ (3) $\sqrt{2g}$ (4) $\sqrt{g}$

3. The work by an ideal monoatomic gas along the cyclic path LMNOL is :



- (1) PV (2) 2PV (3) 3PV (4) 4PV

4. The length of the resistance wire is increased by 10% by stretching. What is the corresponding change in the resistance of wire ?

- (1) 10% (2) 25% (3) 21% (4) 9%

5. The average value of electric energy density in an electromagnetic wave is (E_0 is peak value) :-

- (1) $\frac{1}{2} \epsilon_0 E_0^2$ (2) $\frac{E_0^2}{2\epsilon_0}$
(3) $\epsilon_0 E_0^2$ (4) $\frac{1}{4} \epsilon_0 E_0^2$

6. The quantity $X = \frac{\epsilon_0 L V}{t}$: ϵ_0 is the permittivity of free

space, L is length, V is potential difference and t is time. The dimensions of X are same as that of :-

- (1) Resistance (2) Charge
(3) Voltage (4) Current

7. The velocities of two particles A and B of same mass are $\vec{V}_A = a\hat{i}$ and $\vec{V}_B = b\hat{j}$ where a and b are constants.

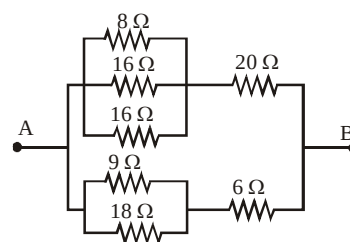
The acceleration of particle A is $(2a\hat{i} + 4b\hat{j})$ and acceleration of particle B is $(a\hat{i} - b\hat{j})$ (in m/s^2). The centre of mass of two particle will move in :-

- (1) straight line (2) parabola (3) ellipse (4) circle

8. A particle of mass m in a unidirectional potential field have potential energy $U(x) = a + 2bx^2$ where a and b are positive constants. Find time period of oscillation :-

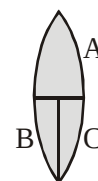
- (1) $2\pi\sqrt{\frac{2b}{m}}$ (2) $2\pi\sqrt{\frac{m}{2b}}$ (3) $\pi\sqrt{\frac{m}{b}}$ (4) $\pi\sqrt{\frac{b}{m}}$

9. The equivalent resistance of the arrangement of resistances shown in adjoining figure between the points A and B is



- (1) 6 ohm (2) 8 ohm (3) 16 ohm (4) 24 ohm

10. A thin, symmetric double convex lens of power P is cut into three parts A , B and C as shown. The power of -

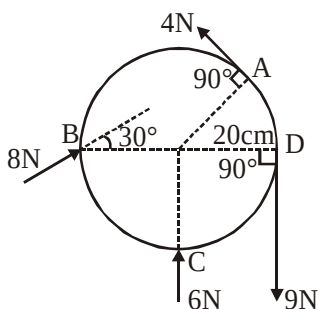


- (1) C is P (2) A is $2P$ (3) B is $\frac{P}{2}$ (4) B is $\frac{3}{2}P$

11. Two cars get closer by 8 m every second while travelling in the opposite directions. They get closer by 0.8 m every second while travelling in the same direction. What are the speeds of the two cars?

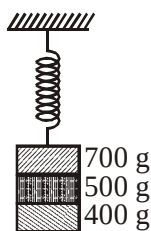
(1) 4 m s^{-1} and 4.4 m s^{-1}
 (2) 4.4 m s^{-1} and 3.6 m s^{-1}
 (3) 4 m s^{-1} and 3.6 m s^{-1}
 (4) 4 m s^{-1} and 3 m s^{-1}

12. A wheel of radius 20 cm has forces applied to it as shown in the figure. The net torque produced by the forces 4N at A, 8N at B, 6N at C and 9N at D at angles indicated is :-



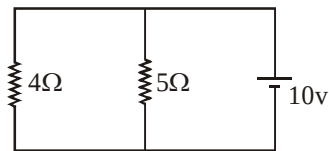
(1) 5.4 N-m anticlockwise
 (2) 1.80 N-m clockwise
 (3) 2.0 N-m clockwise
 (4) 5.4 N-m clockwise

13. Three masses 700 gm, 500 gm and 400 gm are suspended at the end of a spring as shown and are in equilibrium. When the 700 gm mass is removed, the system oscillates with a period of 3 seconds; when the 500 gm mass is also removed, it will oscillate with a period of :



(1) 1 sec (2) 2 sec
 (3) 3 sec (4) $\sqrt{\frac{12}{5}}$ sec

14. In the following diagram current in the 5Ω resistor is (when the internal resistance of the battery is zero) :-



(1) 4.5 A (2) 2A (3) 2.5A (4) zero

15. The distance between the objective lens and the eye lens of an astronomical telescope when adjusted for parallel light is 100 cm. The measured value of the magnification is 19. The focal lengths of the lenses are :-

(1) 85 cm and 15 cm (2) 82 cm and 18 cm
 (3) 95 cm and 5 cm (4) 50 cm and 50 cm

16. Two tall buildings are 30 m apart. The speed with which a ball must be thrown horizontally from a window 150 m above the ground in one building so that it enters a window 27.5 m from the ground in the other building is: (Take $g = 9.8 \text{ m/sec}^2$)

(1) 2 ms^{-1} (2) 6 ms^{-1} (3) 4 ms^{-1} (4) 8 ms^{-1}

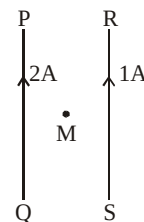
17. A ring and disc roll down an identical inclined plane from same height. The ratio of time taken by them to reach bottom of the inclined plane is :-

(1) $\sqrt{\frac{3}{4}}$ (2) $\sqrt{\frac{4}{3}}$ (3) $\sqrt{\frac{2}{3}}$ (4) $\sqrt{\frac{1}{3}}$

18. A man is standing between two parallel cliffs and fires a gun. If he hears first and second echoes after 1.5 s and 3.5s respectively, the distance between the cliffs is (Velocity of sound in air = 340 ms^{-1})

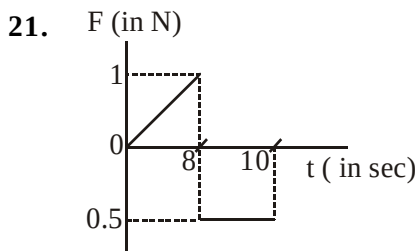
(1) 1190 m (2) 850 m (3) 595 m (4) 510 m

19. PQ and RS are long parallel conductors separated by certain distance. M is the mid-point between them (see the figure). The net magnetic field at M is B. Now, the current 2A is switched off. The field at M now becomes :-



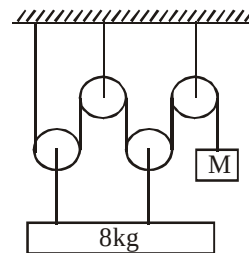
(1) 2B (2) B (3) $\frac{B}{2}$ (4) 3B

20. A light wave is incident normally over a slit of width 24×10^{-5} cm. The angular position of second dark fringe from the central maxima is 30° . What is the wavelength of light :
- (1) 6000 Å (2) 5000 Å
(3) 3000 Å (4) 1500 Å

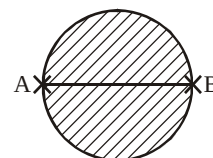


- A force-time graph for the motion of a body is shown in the figure. The change in the momentum of the body between 0 to 10 sec. is :-
- (1) Zero (2) 4 kg m/s
(3) 5 kg m/s (4) 3 kg m/s
22. A research satellite of mass 200 kg circles the earth in an orbit of average radius $3R/2$ where R is the radius of the earth. Assuming the gravitational pull on a mass of 1 kg on the earth's surface to be 10 N, the pull on the satellite will be :-
- (1) 880 N (2) 889 N
(3) 890 N (4) 892 N
23. A glass tube is open at both the ends. A tuning fork of frequency f resonates with the air column inside the tube. Now the tube is placed vertically inside water so that half the length of the tube is filled with water. Now the air column inside the tube is in unison with another fork of frequency f' . Then :-
- (1) $f' = f$ (2) $f' = 4f$
(3) $f' = 2f$ (4) $f' = f/2$
24. A magnet of magnetic moment $50\hat{i}$ A-m² is placed along the x-axis in a magnetic field $\vec{B} = (0.5\hat{i} + 3.0\hat{j})$ T. The torque acting on the magnet is :-
- (1) $175 \hat{k}$ N-m (2) $150 \hat{k}$ N-m
(3) $75 \hat{k}$ N-m (4) $25\sqrt{37} \hat{k}$ N-m

25. The maximum kinetic energy of photoelectrons emitted from a surface when photons of energy 6 eV fall on it is 4 eV. The stopping potential in volts is :-
- (1) 2 (2) 4 (3) 6 (4) 10.
26. What should be the value of mass M so that 8kg block may remain at rest ?



- (1) 2kg (2) 4kg (3) 8kg (4) 16kg
27. Two satellites A and B go around the earth in circular orbits at heights of R_A and R_B respectively from the surface of the earth. Assuming the earth to be a uniform sphere of radius R_e , the ratio of the magnitudes of their orbital velocities is :-
- (1) $\sqrt{\frac{R_B}{R_A}}$ (2) $\frac{R_B + R_e}{R_A + R_e}$
(3) $\sqrt{\frac{R_B + R_e}{R_A + R_e}}$ (4) $\left(\frac{R_A}{R_B}\right)^2$
28. Sphere is charged uniformly. If one moves from A to B along diameter AB, then electric field -



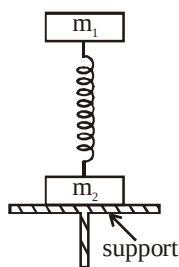
- (1) First increases then decreases.
(2) First decreases then increases.
(3) Remains same.
(4) Zero.
29. The flux linked with a coil at any instant t is given by $\phi = 10 t^2 - 50t + 250$. The induced e.m.f. at $t = 3$ sec is:
- (1) 190 volt (2) -190 volt
(3) -10 volt (4) 10 volt

30. Half life of a radioactive nucleus is 10 years. Probability of decay of a nucleus in 20 years is:-

- (1) 75% (2) 50%
(3) 25% (4) Zero

31. The system of two weights of masses m_1 and m_2 are connected with weight less spring as shown. The system is resting on the support S. If the support S is quickly removed then acceleration of each weight just after the removal of S are :-

[Given $\Rightarrow m_1 = m$ & $m_2 = 2m$]



- (1) $a_1 = g \text{ m/s}^2, a_2 = 2g \text{ m/s}^2$
(2) $a_1 = g/2 \text{ m/s}^2, a_2 = \frac{1}{3}g \text{ m/s}^2$
(3) $a_1 = 0, a_2 = \frac{3}{2}g \text{ m/s}^2$
(4) $a_1 = \frac{2}{3}g \text{ m/s}^2, a_2 = 0$

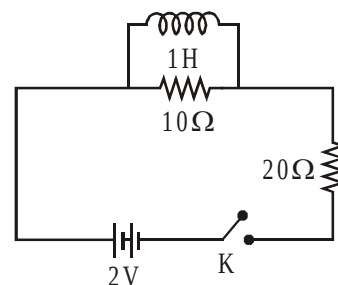
32. A cuboidal piece of wood has dimensions a , b and c . Its relative density is d . It is floating in a large body of water such that side a is vertical. It is pushed down a bit and released. The time period of SHM executed by it is :

- (1) $2\pi\sqrt{\frac{abc}{g}}$ (2) $2\pi\sqrt{\frac{g}{da}}$
(3) $2\pi\sqrt{\frac{bc}{dg}}$ (4) $2\pi\sqrt{\frac{da}{g}}$

33. A small electric dipole is of dipole moment p . The electric potential at a distance ' r ' from its centre and making an angle θ from the axis of dipole will be:-

- (1) $\frac{kp \sin \theta}{r^2}$ (2) $\frac{kp \cos \theta}{r^2}$
(3) $\frac{kp}{r^3} \sqrt{1 + 3 \cos^2 \theta}$ (4) $\frac{kp}{r^3} \sqrt{1 + 3 \sin^2 \theta}$

34. In the adjoining figure what is the final value of current in the 10 ohm resistor when the key K is on?



- (1) $(3/10) \text{ A}$ (2) $(3/20) \text{ A}$
(3) $(3/11) \text{ A}$ (4) Zero

35. X-rays are produced in an X-ray tube operating at a given accelerating voltage. The wavelength of the continuous X-rays has values from :-

- (1) 0 and ∞
(2) $\lambda_{\min}$ to ∞ ; where $\lambda_{\min} > 0$
(3) 0 to $\lambda_{\max}$; where $\lambda_{\max} < \infty$
(4) $\lambda_{\min}$ to $\lambda_{\max}$; where $0 < \lambda_{\min} < \infty$

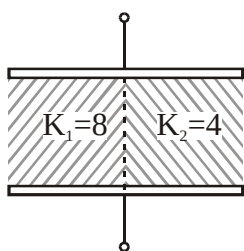
36. A force given by the relation $F = 8t$ acts on a body of mass 2 kg initially at rest. The work done by this force on the body during first 2 s of its motion is :-

- (1) 64 J (2) Zero
(3) 32 J (4) None of these

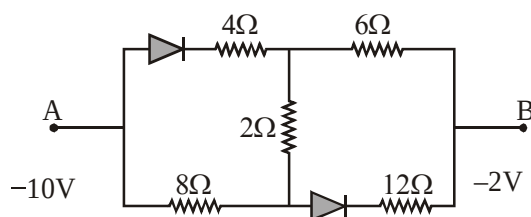
37. Eight drops of a liquid of density ρ and each of radius a are falling through air with a constant velocity 3.75 cms^{-1} . When the eight drops coalesce to form a single drop the terminal velocity of the new drop will be :-

- (1) $15 \times 10^{-2} \text{ ms}^{-1}$ (2) $2.4 \times 10^{-2} \text{ ms}^{-1}$
(3) $0.75 \times 10^{-2} \text{ ms}^{-1}$ (4) $25 \times 10^{-2} \text{ ms}^{-1}$

38. A capacitor having capacitance $1\mu\text{F}$ with air is filled with two dielectrics as shown. How many times capacitance will increase?

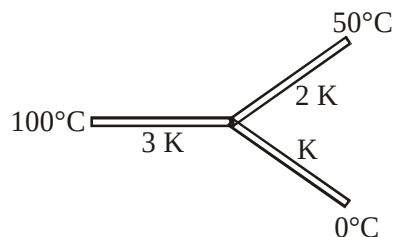


- (1) 12 (2) 6
(3) $8/3$ (4) 3
39. The r.m.s. current in an ac circuit is 2 A. If the wattless current be $\sqrt{3}$ A, what is the power factor:-
- (1) $\frac{1}{\sqrt{3}}$ (2) $\frac{1}{\sqrt{2}}$
(3) $\frac{1}{2}$ (4) $\frac{1}{3}$
40. In the following circuit the equivalent resistance between A and B is :-

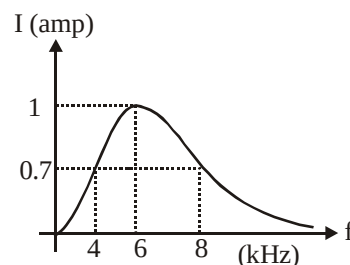


- (1) $\frac{20}{3}\Omega$ (2) 10Ω
(3) 16Ω (4) 20Ω
41. Velocity of a body of mass 2 kg moving in xy plane is given by $\vec{v} = (2\hat{i} + 4t\hat{j})$ m/s, where t is the time in second. The power delivered to the body by the resultant force acting on it at $t = 5$ sec. is
- (1) 80 W (2) 160 W
(3) 40 W (4) 100 W

42. Three rods of the same dimensions have thermal conductivities 3K, 2K and K. They are arranged as shown, with their ends at 100°C , 50°C and 0°C . The temperature of their junction is :-



- (1) 75°C (2) $\frac{200}{3}^\circ\text{C}$ (3) 40° (4) $\frac{100}{3}^\circ\text{C}$
43. If dielectric constant and dielectric strength be denoted by K and X respectively then a material suitable for use as dielectric in a capacitor must have:-
(1) high K and high X (2) high K and low X
(3) low K and high X (4) low K and low X
44. The plot given below is current delivered to an LRC circuit versus frequency. The quality factor of the circuit is :



- (1) 5 (2) 2.4 (3) 1.5 (4) 1.2
45. In the following combinations of logic gates, the outputs of A, B and C are respectively :-
- (i)
- (ii)
- (iii)
- (1) 0, 1, 1 (2) 0, 1, 0 (3) 1, 1, 0 (4) 1, 0, 1

46. Incorrect regarding polythionic series $H_2S_nO_6$ is :-
 (1) If value of $n \uparrow$ average oxidation state of sulphur $\downarrow$
 (2) If value of $n = 4$, no. of s-s linkage = 3
 (3) If value of $n \uparrow$ or $\downarrow$ no. of = bond remain constant
 (4) If value of $n \uparrow$ no. of sulphur contain + 5 oxidation state $\downarrow$
47. $Ca(OH)_2 + Cl_2 \rightarrow C + H_2O$
 Oxidation state of Cl in 'x'
 (1) +1 only (2) +1 & +2
 (3) +1 & -1 (4) -1 only
48. Which of the following statements is wrong ?
 (1) Ethyl chloride on reduction with Zn-Cu couple and alcohol gives ethane
 (2) The reaction of methyl magnesium bromide with acetone gives butanol-2
 (3) Hydrogen halides follow the following reactivity sequence on reaction with alkenes
 $H-I > H-Br > H-Cl > H-F$
 (4) $C_2H_4Cl_2$ may exist in two isomeric forms.
49. Which of the following cannot be considered as a step of mechanism in chain reaction of methane with Cl_2 ?
 (1) $Cl_2 \rightarrow 2\dot{Cl}$
 (2) $CH_4 + Cl\cdot \rightarrow CH_3\cdot + HCl$
 (3) $Cl\cdot + CH_4 \rightarrow \dot{C}H_3 + HCl$
 (4) $Cl\cdot + \dot{C}H_3 \rightarrow CH_3\cdot + Cl$
50. The difference between the wave number of 1st line of Balmer series and last line of Paschen series for Li^{2+} ion is:-
 (1) $\frac{R}{36}$ (2) $\frac{5R}{36}$ (3) $4R$ (4) $\frac{R}{4}$
51. Which of the following Halogen doesn't form any oxide ?
 (1) F (2) Cl (3) Br (4) I
52. Which of the following oxide of Halogen used in a paper industry as well as water purification?
 (1) I_2O_5 (2) ClO_2
 (3) I_2O_4 (4) Cl_2O_7

53. Match column-I (reaction) with column-II (type of reaction) and select the correct option from the codes given below :-

Column-I (Reaction)	Column-II (Type of reaction)
(A)	(X) Electrophilic addition
(B)	(Y) Electrophilic substitution
(C)	(Z) Nucleophilic addition
(D)	(W) Free radical substitution

Codes :-

A	B	C	D
(1) X	Y	Z	W
(2) W	Z	X	Y
(3) Z	W	X	Y
(4) W	Z	Y	X

54. Addition of HCl to 2-methyl-2-butene mainly gives.
 (1) 1-chloro-2-methyl butane
 (2) 2-chloro-2-methyl butane
 (3) 2-chlorobutane
 (4) 1-chlorobutane
55. $\log \frac{K_p}{K_c} + \log RT = 0$ is a relationship for the gaseous reaction
 (1) $PCl_5 \rightleftharpoons PCl_3 + Cl_2$ (2) $2SO_2 + O_2 \rightleftharpoons 2SO_3$
 (3) $H_2 + I_2 \rightleftharpoons 2HI$ (4) $N_2 + 3H_2 \rightleftharpoons 2NH_3$
56. In a following reaction
 $XeF_n + CsF \longrightarrow Cs^+ [XeF_{n+1}]^-$
 Which value of 'n' having maximum tendency to takes place this reaction.
 (1) 2 (2) 4
 (3) 6 (4) Can't be predicted

57. Which of the following is used to enrichment of uranium ?

- (1) ClF_3 (2) ICl
(3) IF_7 (4) F_2

58. The correct match between Item-I and Item-II is:-

Item-I (Drug)		Item-II (Test)	
(A)	Chloroxylenol	(P)	Carbylamine test
(B)	Norethindrone	(Q)	Sodium hydrogen carbonate test
(C)	Sulphapyridine	(R)	Ferric chloride test
(D)	Penicillin	(S)	Bayer's test

Codes :-

- | A | B | C | D |
|-------|---|---|---|
| (1) R | P | S | Q |
| (2) Q | S | P | R |
| (3) R | S | P | Q |
| (4) Q | P | S | R |

59. Select the correct reactivity order of dehydrohalogenation reaction for the following halides with alcoholic KOH :-

- (i) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{Br}$
(ii) $\text{CH}_3\text{CH}(\text{Br})\text{CH}(\text{CH}_3)_2$
(iii) $\text{CH}_3\text{CH}(\text{Br})\text{CH}_2\text{Ph}$

- (1) (i) > (ii) > (iii)
(2) (i) > (iii) > (ii)
(3) (iii) > (ii) > (i)
(4) (ii) > (iii) > (i)

60. Which of following process is not responsible for the presence of charge on surface of sol particles?

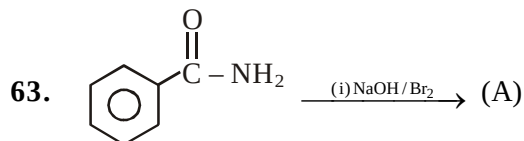
- (1) Electron capture by sol particle.
(2) Adsorption of ionic species from solution.
(3) Formation of Helmholtz electrical double layer.
(4) Absorption of ionic species from solution.

61. Which of the following is not an example of Redox reaction ?

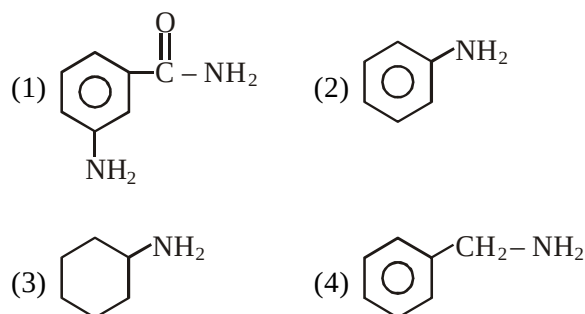
- (1) $(\text{NH}_4)_2\text{Cr}_2\text{O}_7 \xrightarrow{\Delta}$
(2) $\text{XeF}_2 + \text{H}_2\text{O} \longrightarrow$
(3) $\text{AgNO}_3 \xrightarrow{\Delta}$
(4) $\text{XeF}_6 + \text{H}_2\text{O} \longrightarrow$

62. Froth floatation process for the concentration of sulphide ores is an illustration of the practical application of

- (1) adsorption (2) Absorption
(3) Bessemerisation (4) Coagulation



Product A is :-



64. At identical temperature and pressure, the rate of diffusion of hydrogen gas is $3\sqrt{3}$ times that of a gaseous hydrocarbon having molecular formula $\text{C}_n\text{H}_{2n-2}$. What is the value of n ?

- (1) 1 (2) 4 (3) 3 (4) 8

65. The combustion enthalpy of carbon, hydrogen and methane are -393.5 ; -284.8 and -890.4 kJ/mole respectively at 25°C . The value of standard formation enthalpies of methane at that temperature is :-

- (1) 890.4 kJ/mole
(2) -298.8 kJ/mole
(3) -72.7 kJ/mole
(4) -107.7 kJ/mole

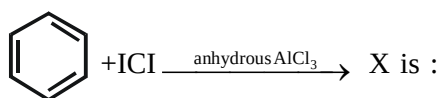
66. During change of N_2 to N_2^- ion, the e^- add on which one of the following orbital.

- (1) σ^* orbital (2) σ orbital
(3) π^* orbital (4) π -orbital

67. EDTA can acts as ?

- (1) Complexing agent
(2) Chelating ligand
(3) Antidote for lead poisoning
(4) All of the above

68. The compound X in the reaction,



- (1) (2)
 (3) (4)

69. Find work done during expansion of a gas from a volume of 4dm^3 to 6dm^3 against a constant external pressure of 3 atm is :-

- (1) -304 J (2) -6 J
 (3) -608 J (4) $+304\text{ J}$

70. In the reaction $4\text{Fe} + 3\text{O}_2 \rightleftharpoons 4\text{Fe}^{+3} + 6\text{O}^{--2}$, which of the following statement is incorrect ?

- (1) Metallic iron is a reducing agent
 (2) Fe^{+3} is an oxidising agent
 (3) It is disproportionation reaction
 (4) Fe^{+3} is oxidised form of metallic iron

71. Boron + Conc. $\text{HNO}_3 \rightarrow \text{H}_3\text{BO}_3$ + gas 'a' + H_2O
 Boron + Conc. $\text{H}_2\text{SO}_4 \rightarrow \text{H}_3\text{BO}_3$ + gas 'b' + H_2O
 Hybridisation of gas 'a' & 'b' are respectively.

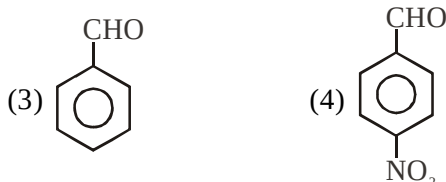
- (1) sp^2 & sp^2 (2) sp & sp^2
 (3) sp^2 & sp^3 (4) sp^3 & sp^3

72. Chromyl chloride test doesn't gives by Cl^- of

- (1) Ag^+ (2) Na^+
 (3) Ca^{+2} (4) K^+

73. Which of the following compound not give Cannizaro reaction ?

- (1) HCHO (2) $\text{CH}_3\text{-CHO}$



74. Number of moles of hydrogen atoms present in 60 amu of acetic acid :

- (1) $4 / N_A$ (2) 4 (3) 1 (4) $1 / N_A$

75. How many unit cells are present in a cube shaped ideal crystal of NaCl of mass 1 gram

(molar mass of NaCl = 58.5 g) :-

- (1) 2.57×10^{21} (2) 8×10^{21}
 (3) 1.7×10^{21} (4) 5×10^{21}

76. $\text{CH}_2(\text{COOH})_2 \xrightarrow{\text{P}_4\text{O}_{10}/\Delta} \text{'X'} + \text{H}_2\text{O}$ 'X' doesn't contain.

- (1) $\text{p}\pi\text{-p}\pi$ bond
 (2) All 'C' having same hybridisation
 (3) Total 4 double bond
 (4) no. of σ to π bond ratio is 1.5

77. (i) $\text{CH}_2 = \text{CH} - \text{CH}_2 - \text{CH} = \text{CH}_2$

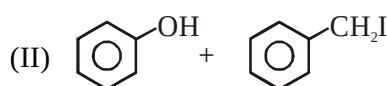
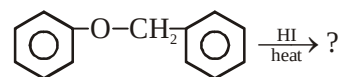
(ii) $\text{CH}_2 = \text{CH} - \text{CH} = \text{CH} - \text{CH}_3$

(iii) $\text{CH}_3 - \text{CH} = \text{CH} - \text{CH} = \text{CH} - \text{CH}_3$

The number of possible geometrical isomers for the above compounds respectively are.

- (1) 0,2,4 (2) 2,2,4 (3) 0,3,3 (4) 0,2,3

78. What are the major products in the following reaction?



- (1) I (2) II (3) III (4) IV

79. When NH_4Cl salt is dissolved in water it gets hydrolysed and forms weak base and strong acid. Find equilibrium constant of the reaction. If ionisation constant of weak base is 10^{-4} :

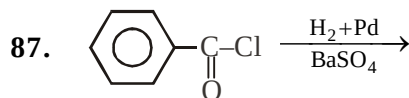
- (1) 10^{-4} (2) 2×10^{-4}
 (3) 10^{-10} (4) 2×10^{-10}

80. Which type of structure does RbCl have if ionic radii of Rb^+ and I^- respectively are 1.46 Å and 2.16 Å:-
 (1) NaCl type (2) CsCl type
 (3) ZnS type (4) CaF_2 type
81. Which of the following pair of species are high spin complex ?
 (1) $\text{K}_3[\text{Mn}(\text{CN})_6]$ & $[\text{Cr}(\text{NH}_3)_6]\text{Cl}_3$
 (2) $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$ & $[\text{Ni}(\text{NH}_3)_6]^{2+}$
 (3) $\text{K}_3[\text{Fe}(\text{CN})_6]$ & $\text{K}_4[\text{Fe}(\text{CN})_6]$
 (4) $[\text{CoF}_6]^{3-}$ & $[\text{CrF}_6]^{3-}$
82. Bottles containing $\text{C}_6\text{H}_5\text{I}$ and $\text{C}_6\text{H}_5\text{CH}_2\text{I}$ lost their original labels. They were labelled A and B for testing. A and B were separately taken in test tubes and boiled with NaOH solution. The end solution in each tube was made acidic with dilute HNO_3 and then some AgNO_3 solution was added. Substance B gave a yellow precipitate. Which one of the following statements is true for this experiment ?
 (1) A is $\text{C}_6\text{H}_5\text{I}$
 (2) A is $\text{C}_6\text{H}_5\text{CH}_2\text{I}$
 (3) B is $\text{C}_6\text{H}_5\text{I}$
 (4) Addition of HNO_3 is unnecessary
83. The incorrect statement among the following is :-
 (1) α -D-glucose and β -D-glucose are anomers
 (2) α -D-glucose and β -D-glucose are enantiomers
 (3) Cellulose is a straight chain polysaccharide made up of only β -D-glucose
 (4) The penta acetate of glucose does not react with hydroxyl amine
84. Vapour pressure of solution of benzene ($P^\circ = 120$ torr) and toluene ($P^\circ = 80$ torr) having 2 moles of each is:-
 (1) 400 torr (2) 200 torr
 (3) 100 torr (4) 50 torr
85. What is the order of reaction for $\text{A} + \text{B} \rightarrow \text{C}$.

exp.	[A]	[B]	Rate of Reaction
1	0.1M	0.1M	$2.0 \times 10^{-3} \text{ mol L}^{-1} \text{ s}^{-1}$
2	0.4M	0.1M	$0.4 \times 10^{-2} \text{ mol L}^{-1} \text{ s}^{-1}$
3	0.1M	0.2M	$1.4 \times 10^{-2} \text{ mol L}^{-1} \text{ s}^{-1}$

- (1) 1 (2) 3
 (3) 3.33 (4) 4

86. Incorrect regarding $\text{Xe}[\text{PtF}_6]$ & $\text{O}_2[\text{PtF}_6]$
 (1) Pt in +5 oxidation state
 (2) Xe & O_2 having nearly equal Ionisation energy
 (3) O-O bond length in O_2 is more than $\text{O}_2[\text{PtF}_6]$
 (4) In $\text{O}_2[\text{PtF}_6]$, O_2 contain -1 charge



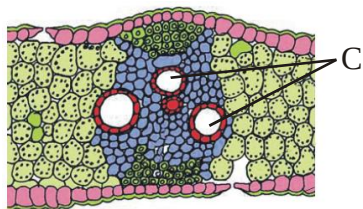
The above reaction is known as :

- (1) Curtius reaction (2) Rosenmund's reduction
 (3) Mendius reduction (4) Stephen's reaction
88. Match the polymers in column-I with their uses in column-II and choose the correct answer.

Column-I		Column-II	
(A)	Polystyrene	(i)	Paints and lacquers
(B)	Glyptal	(ii)	Rain coats
(C)	Polyvinyl chloride	(iii)	Manufacture of toys
(D)	Bakelite	(iv)	Computer discs

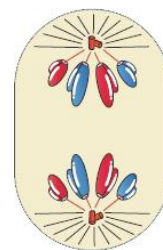
- | | | | |
|-----------|----------|----------|----------|
| A | B | C | D |
| (1) (ii) | (i) | (iii) | (iv) |
| (2) (ii) | (iv) | (iii) | (i) |
| (3) (iii) | (iv) | (ii) | (i) |
| (4) (iii) | (i) | (ii) | (iv) |
89. The decomposition of a hydrocarbon follows the equation $k = (4.5 \times 10^{11} \text{ s}^{-1})e^{-28000 \text{ K/T}}$. What will be the value of activation energy ?
 (1) 669 kJ mol^{-1} (2) 232.79 kJ mol^{-1}
 (3) 4.5×10^{11} kJ mol^{-1} (4) 28000 kJ mol^{-1}
90. Which of the following statement is incorrect?
 (1) The conductivity of metal is due to mobility of electrons.
 (2) Specific conductance is the conductance of 1 cm^3 electrolyte solution.
 (3) Specific conductance increases while equivalent conductivity decreases on dilution.
 (4) The maximum value of molar conductivity for weak electrolyte can not be determined by extrapolation of curve in between $\wedge_m$ and $\sqrt{C}$.

91. Stored food in brown algae is/are :-
 (1) Glycogen and oil
 (2) Laminarin and mannitol
 (3) Floridean starch
 (4) Cellulose
92. Given figure represent internal structure of mono cotyledonous leaf, in which labelled part 'C' is :-



- (1) Mesophyll (2) Phloem
 (3) Xylem (4) Sub-Stomatal cavity
93. 40s and 60s are actively synthesized in:-
 (1) Nucleolus (2) Ribosome
 (3) Lysosome (4) Nucleosome
94. Acetyl Co-A then enters the tricarboxylic acid pathway or kreb's cycle operating in the matrix of mitochondria then:-
 (1) $\text{NADH} + \text{H}^+$, FADH_2 and O_2 are produced
 (2) $\text{NADH} + \text{H}^+$, O_2 and H are released
 (3) FADH_2 , ATP and O_2 are produced
 (4) $\text{NADH} + \text{H}^+$, FADH_2 and ATP are produced
95. Fill in the blanks.
 A is a unit of land with a natural boundary having a mosaic of patches. These patches generally represent different
 (1) Biome, Landscape
 (2) Biome, Ecosystem
 (3) Landscape, Ecosystem
 (4) Landscape, Biome
96. In galapagos island, Darwin found different varieties of finches, originated from a common ancestor. The original variety of these finches was:-
 (1) Insectivorous (2) Vegetarian
 (3) Seed-eating (4) Cactus eating

97. Site of fertilization in human female is :-
 (1) Infundibulum of fallopian tube
 (2) Isthmus of fallopian tube
 (3) Ampulla of fallopian tube
 (4) Intra mural part of fallopian tube
98. Which of following restriction enzymes produces blunt ends ?
 (1) BamHI (2) Hind III (3) Sal I (4) EcoRV
99. Renal corpuscles are formed by :-
 (1) Arteriole and glomerulus
 (2) Afferent and efferent arterioles
 (3) Arterioles and bowman's capsule
 (4) Bowman's capsule and glomerulus
100. Which hormone is secreted during fall in blood pressure:-
 (1) Oxytocin (2) Atrial natriuretic factor
 (3) Aldosterone (4) androgen & Oestrogen
101. In the sexual reproduction of red algae the gametes are :-
 (1) Biflagellate (2) Motile
 (3) Multiflagellate (4) Non motile
102. In which of the following pericycle is present on the inner side of the endodermis and above the phloem in the form of semi-lunar patches of sclerenchyma?
 (1) *Zea mays* root (2) *Helianthus annuus* stem
 (3) *Zea mays* stem (4) *Cucurbita pepo* stem
103. A stage in mitosis is shown in the figure select the answer which gives correct identification of the stage:-



- (1) Anaphase (2) Prophase
 (3) Metaphase (4) Telophase
104. Which one plant hormone plays an important role in seed development, maturation and dormancy?
 (1) Auxins (2) Gibberellins
 (3) Ethylene (4) Absciscic Acid

105. In ecological succession, the individual transitional communities are termed :-
 (1) Sere (2) Pioneer Community
 (3) Seral community (4) Climax community
106. If mouth is present and anus is absent. This type of body plan is called :-
 (1) Cell aggregate body plan
 (2) Blind sac body plan
 (3) Tube with in tube body plan
 (4) Tissue body plan
107. Time required for completion of spermatogenesis in human being is :-
 (1) 10-15 days (2) 65-70 days
 (3) 6-8 months (4) 10-14 years
108. Which of the following enzymes are used to obtain protoplast ?
 (1) Cellulase and pectinase (2) Cellulase and protease
 (3) Chitinase and cellulase (4) Lipase and cellulase
109. Among the following how many conditions are favourable for dissociation of oxygen from oxyhaemoglobin :-
 (i) Low pO_2 (ii) High pCO_2
 (iii) High pH (iv) Low temperature
 (v) High H^+ concentration
 (1) 3 (2) 5 (3) 4 (4) 2
110. How many statements given below are **wrong**?
 (A) With repeated use of drugs, the tolerance level of the receptors present in our body increases.
 (B) Smoking increases carbon monoxide (CO) content in the blood and reduces the concentration of haembound oxygen.
 (C) Cocaine is chemically diacetylmorphine.
 (D) The plant illustrated is *Atropa belladonna* and has hallucinogenic properties.



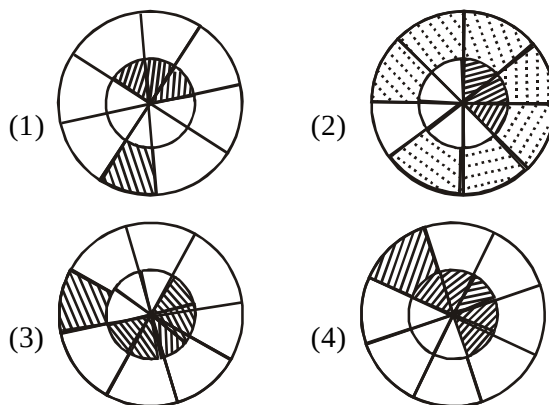
Options

- (1) One (2) Two (3) Three (4) Four

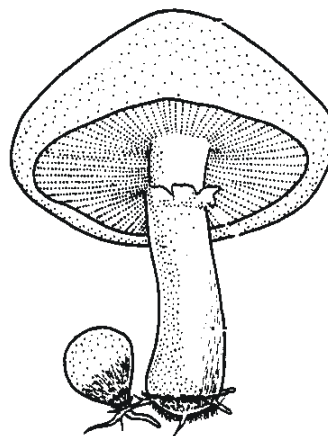
111. Unbranched stem is character of which gymnosperm:-
 (1) *Pinus* (2) *Cycas*
 (3) *Cedrus* (4) *Lycopodium*
112. In the dicot root, the vascular cambium is :-
 (1) Partly primary and partly secondary in origin
 (2) Completely primary in origin
 (3) Completely secondary in origin
 (4) Only partly primary in origin
113. Which of the following statement is not correct ?
 (1) Chromatin contains DNA, RNA, histone and non-histone protein
 (2) Every chromosome has a primary constriction
 (3) Centrosome holds two chromatids of a chromosome
 (4) In acrocentric chromosome the centromere is situated close to its end
114. Which one is monocarpic and biennial plant:-
 (1) Wheat (2) rye (3) Cabbage (4) Barley
115. k-selected species has :-
 (1) High stability & low biotic potential
 (2) Low stability & high biotic potential
 (3) Short life time & high growth rate
 (4) Long life time & high growth rate
116. Pick out the odd one :-
 (1) Meandrina _____ brain coral
 (2) Pennatula _____ sea pen
 (3) Sycon _____ bath sponge
 (4) Gorgonia _____ sea fan
117. Which one of the following is not include in natural method of contraception :-
 (1) Periodic abstinence
 (2) Withdrawal
 (3) Lactational amenorrhoea
 (4) Medical termination of pregnancy
118. Read the following statements carefully :-
 (A) Wheat variety, Atlas-66 having a high protein content
 (B) SCP is the alternate source of proteins for animal and human nutrition
 (C) Plants developed by micropropagation will be genetically different to the original plant from which they were grown
 (D) Semi-dwarf rice varieties was derived from IR-8 Taichung Native-I
 Find out the correct statement :-
 (1) ABC (2) CDA (2) BCD (4) ABD

119. In human, cell fragments produced from Megakaryocytes are called :-
 (1) Lymphocytes (2) Erythrocytes
 (3) Thrombocytes (4) Lecucocytes
120. Rheumatoid arthritis is a type of :-
 (1) Allergy (2) Autoimmune disease
 (3) Syndrome (4) Bone cancer
121. Severe famine of west Bengal 1942—1943 was due to the destruction of rice crop by a fungus called :-
 (1) *Helminthosporium* (2) *Penicillium*
 (3) *Rhizopus* (4) *Puccinia*
122. Secondary medullary rays passes through which part of bark in the radial directions:-
 (1) Phellem (2) Phellogen
 (3) Phelloderm (4) Secondary phloem
123. Which one is primary metabolite?
 (1) Anthocyanin (2) Chlorophylls
 (3) Abrin (4) Rubber
124. α -Ketoglutaric Acid + NH_4^+ + NADPH $\xrightarrow[\text{Dehydrogenase}]{\text{Glutamate}}$ glutamate + H_2O + NADP
 Above given reaction represents:-
 (1) Transamination
 (2) Deamination
 (3) Reductive amination
 (4) Nitrification
125. Biological equilibrium is an equilibrium amongst?
 (1) Producer and decomposer
 (2) Producer and consumers
 (3) Producer, consumers and decomposers
 (4) Consumer and decomposer
126. How many malpighian tubules present at the junction of mid-gut and hind-gut in cockroach :-
 (1) 60-100 (2) 90-110
 (3) 100-150 (4) 70-90
127. Amniocentesis technique is not use for detection of:-
 (1) Genetic disorder (2) Metabolic disorder
 (3) Barr bodies (4) Heart disease
128. Ploidy of cells of the tetrad is :-
 (1) n (2) 2n
 (3) 3n (4) 4n

129. Mark the correct figure which represent cardiac cycle:-



130. The four polypeptide chains of the antibody molecule are held together by :-
 (1) Peptide bonds
 (2) Disulphide bonds
 (3) Ionic bonds
 (4) Vanderwaal interactions
131. Above diagram belongs to which class :-

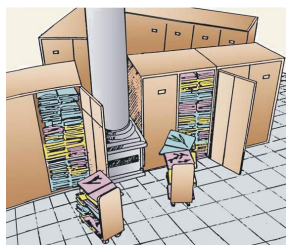


- (1) Ascomycetes (2) Basidiomycetes
 (3) Zygomycetes (4) Oomycetes
132. Detoxification of drugs, pollutants and steroids occurs in:-
 (1) RER (2) Golgi complex
 (3) Glyoxysome (4) SER
133. Choose the correct option for fungal cellulose:-
 (1) Linear polymer of β -D Glucose units
 (2) Main stored food in plants
 (3) Second most abundant organic molecules on earth.
 (4) Anticoagulant of blood

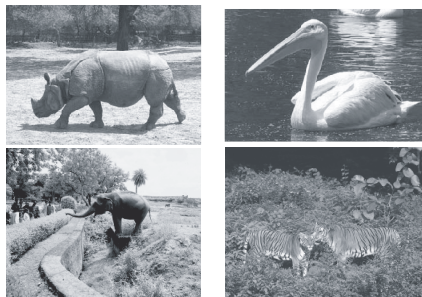
- 134.** In inner membrane of mitochondria, electrons transfer from NADH to oxygen is an :-
 (1) Endergonic process (2) Uphill process
 (3) Exergonic process (4) Reduction of NADH
- 135.** Intergovernmental panel on climate change (IPCC) periodically makes an assesment of the atmospheric abundance of :-
 (1) Only O.D.S. (2) Only S.P.M.
 (3) Only VOCs (4) Green house gases
- 136.** Ureters act as urinogenital ducts in :-
 (1) both male and females frog
 (2) Frog's males
 (3) Human males
 (4) Human female
- 137.** The percentage of abc gamete produced by AaBbCc parent will be :-
 (1) 12.5 (2) 25 (3) 50 (4) 75
- 138.** Which of the following is a dioecious plant ?
 (1) Castor (2) Maize
 (3) Papaya (4) Cucurbits
- 139.** 
- The above diagram is an example of :-
 (1) Inbreeding
 (2) Out-crossing
 (3) Cross-breeding
 (4) Interspecific hybridisation
- 140.** Schick test is used for the investigation of :
 (1) Tuberculosis (2) pertussis
 (3) Leprosy (4) Diphtheria
- 141.** Which fungi causes, white rust disease of crucifers?
 (1) *Rhizopus* (2) *Mucor*
 (3) *Albugo* (4) *Phythium*
- 142.** Which function is related with mitochondria?
 (1) Carbohydrate synthesis
 (2) Photo phosphorylation
 (3) Oxidative phosphorylation
 (4) Basal body formation

- 143.** Choose the correct option for glycerol?
 (1) Used to form Rayon fibres
 (2) Also called as plant cement
 (3) Also called as animal starch
 (4) Participate in the composition of a fat molecules.
- 144.** In which of the following, the vomiting centre is situated?
 (1) Cerebrum (2) Cerebellum
 (3) Medulla oblongata (4) Hypothalamus
- 145.** Problems that have come in the wake of the green revolution :-
 (1) Soil erosion and Acidity
 (2) Lack of CO₂ in Atmosphere
 (3) Water logging and salinity
 (4) Excessive growth of crop plant
- 146.** Cells responsible for secretion of testosterone in male testis is :-
 (1) Sertoli cell
 (2) Leydig's cell
 (3) Spermatogonia
 (4) Immunological cells
- 147.** At a particular locus, frequency of A allele in 0.4 and that of a is 0.6. What would be the frequency of dominant individuals in a random population at equilibrium :-
 (1) 0.24 (2) 0.16
 (3) 0.64 (4) 0.36
- 148.** Manubrium is part of :-
 (1) Scapula (2) Calvicle
 (3) Sternum (4) Humerus
- 149.** Consider the following animals.
 Cow, Sheep, Rabbits, Buffaloes, Mares, Hen, Turkey, Geese.
 How many above animals can be improved by MOET (Multiple Ovulation Embryo Transfer Technology) :-
 (1) 8 (2) 6 (3) 4 (4) 5
- 150.** Killed bacteria is present in which of the following vaccine :
 (1) BCG (2) OPV
 (3) Cholera vaccine (4) Small pox vaccine

151.



(A)



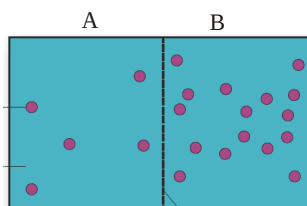
(B)

First of all identify the A and B then their correct function and they are made for which type of organisms. Which one of these options is most correct with respect to identification, uses and about the organisms :-

		Identification	Uses	Made for which organisms
(1)	A	Herbarium	For taxonomy of plants only	Plant species
	B	Zoological park	For taxonomy of animals only	Animal species
(2)	A	Zoological park	For taxonomy of animals	Animal species
	B	Herbarium	For taxonomy of plants	Plant species
(3)	A	Herbarium	For taxonomy of both plants and animals	For both plants and animals
	B	Zoological park	For taxonomy of both plants and animals	For both plants and animals
(4)	A	Museum	For taxonomy of both plants and animals	For both plants and animals
	B	Zoological park	For taxonomy of both plants and animals	For both plants and animals

152. Which of the following events does not occur in Golgi complex ?

- (1) Formation of acrosome
- (2) Formation of lysosome
- (3) Synthesis of glycoprotein and glycolipid.
- (4) Synthesis of protein and lipid



153.

If in chamber A solute molecules are increase than chamber B, then choose the correct option for given figure:-

- (1) Water potential of chamber A will increases
- (2) OP of chamber B is more than chamber A
- (3) DP due to water molecules in chamber A is more than chamber B
- (4) Water will Net move from B to A chamber

154. The dorsal root of spinal cord contains:-

- (1) Somatic motor fibres
- (2) Visceral motor fibres
- (3) Somatic sensory fibres
- (4) All of the above

155. Match the following-

Column-I (Type of interaction)	Column-II (Organism)
a. Proto-cooperation	(i) Parthenium and other angiosperm
b. Commensalism	(ii) Sea anemone and clown fish
c. Amensalism	(iii) Cuckoo and crow
d. Parasitism	(iv) Hermit crab and sea anemone
(1) a-ii, b-i, c-iv, d-iii	(2) a-iii, b-ii, c-iv, d-i
(3) a-iv, b-ii, c-i, d-iii	(4) a-i, b-ii, c-iii, d-iv

156. Theca layer is formed at which stage of follicular development in female ?

- (1) Primary follicle
- (2) Secondary follicle
- (3) Tertiary follicle
- (4) Graafian follicle

157. On an average, how many purine N_2 bases are present in single coil of DNA :-

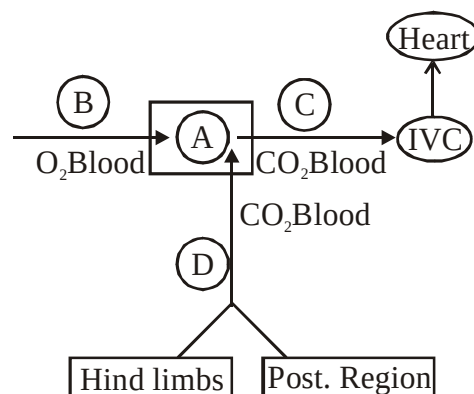
- (1) Four
- (2) Five
- (3) Ten
- (4) Uncertain

158. Which process does not occur during muscle contraction ?
 (1) Actin filament sliding on myosin filament
 (2) Formation of cross bridge
 (3) Myosin filament sliding on actin filament
 (4) ATP hydrolysis
159. Which of the following cell is not abundant in loose connective tissue?
 (1) Fibroblast (2) Lymphocyte
 (3) Macrophage (4) Mast cell
160. Underground stem modified for food storage having nodes, internodes and scaly leaves and growing vertically is called :-
 (1) Corm (2) Bulb (3) Rhizome (4) Tuber
161. Read the following paragraph and fill in the blanks with appropriate words :-
 In pteridophytes the main plant body is ____ (A) ____ which bears sporangia that are subtended by leaf-like appendages called ____ (B) _____. The sporangia produce spores by ____ (C) ____ and after germination of spores, they give rise to photosynthetic gametophytes called ____ (D) _____.
 Choose the correct option :-

	A	B	C	D
(1)	Sporophyte	Sporangia	Meiosis	Protonema
(2)	Gametophyte	Sporophylls	Mitosis	Sporophyte
(3)	Sporophyte	Sporophylls	Meiosis	Prothallus
(4)	Gametophyte	Sporophylls	Mitosis	Protonema

162. Microtubules are not constituents of :-
 (1) Basal body and spindle fibres
 (2) Cilia and flagella
 (3) Centriole and spindle fibres
 (4) Nucleosome and polysome
163. (A) Membrane functioning
 (B) Cell elongation
 (C) Pollen germination
 (D) Maintain the anion-cation Balance of cell
 (E) constituent of ring structure of chlorophyll.
 Choose the option given below which have all the correct function of Boron and Potassium :-
 (1) Boron-A,B,E, potassium -C,D
 (2) Boron B,C,E Pottassium-A,D
 (3) Boron -A,B,C, Potassium-D
 (4) Boron-A,B,C, Pottassium-D,E

164. Glaucoma is an eye disease arising from:-
 (1) Increased pressure of fluid in eye ball
 (2) Elongation of eye ball
 (3) Shortening of eye ball
 (4) Irregularity in the surface of cornea
165. Excess use of herbicides, pesticides has only resulted in selection of resist varieties in a much lesser time scale. It was an example of evolution by :-
 (1) Mass extinction (2) Anthropogenic action
 (3) Continental drift (4) Both (1) and (3)
166. At which stage of embryonic development, implantation occur ?
 (1) Morula (2) Gastrula
 (3) Blastula (4) Zygote
167. Which of the following codon are stop codon :-
 (1) UAA, UAG, UGA (2) AUG, GUG, GAG
 (3) UAA, UAG, UCA (4) AAA, CCC, GGG
168. Given below a sketchic diagram of blood circulation in specific region of frog.



- In above figure what are the A, B, C and D and select the correct option :-
 (1) A = Kidney, B = Renal vein, C = Renal artery, D = Renal portal vein
 (2) A = Liver, B = Hepatic artery, C = Hepatic vein, D = Hepatic portal vein
 (3) A = Kidney, B = Renal artery, C = Renal vein, D = Renal portal vein
 (4) A = Kidney, B = Renal artery, C = Renal portal vein, D = Renal vein

169. Junction which prevents leakage of substances across tissue is:-

- (1) Desmosome
- (2) Tight junction
- (3) Gap junction
- (4) 1 & 2 both

170. In which inflorescence flowers are produced in basipetal order :-

- (1) Racemose
- (2) Cymose
- (3) Solitary
- (4) Cyathium

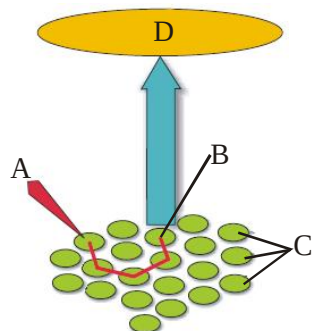
171. Select erroneous pair:-

- (1) Ploidy of the meristem cell of monocot – Diploid
- (2) Primary xylem is endarch – Sunflower stem
- (3) Obliterated central lumen– Xylem fibre
- (4) Phellogen – Primary meristem

172. Cellular organelles with single membrane are :-

- (1) Ribosome and Centrosome
- (2) Mitochondria and Ribosome
- (3) Lysosome and peroxysome
- (4) Chloroplast and Mitochondria

173.



Which one option is incorrect with the respect of given figure (A,B,C and D):-

- (1) A is responsible for excitement of C and B
- (2) C are antennae molecules
- (3) C are directly donate their electrons to D
- (4) D obtain electrons from B

174. What is the intensity of sound in normal conversation?

- (1) 10–20dB
- (2) 35–60dB
- (3) 200-300dB
- (4) 120–150dB

175. We are still not very clear that evolution is a process or the result of a process. But when we describe the story of this world we describe that evolution is a :-

- (1) Process
- (2) Consequence of natural selection
- (3) Non known
- (4) Both (1) and (2)

176. From earliest to latest, the overall sequence of early development proceeds as follow :-

- (1) Cortical reaction → synthesis of embryo's DNA begins → Acrosomal reaction → first cell division
- (2) Synthesis of embryo's DNA begins → cortical reaction → acrosomal reaction → first cell division
- (3) First cell division → acrosomal reaction → cortical reaction → synthesis of embryo's DNA begins
- (4) Acrosomal reaction → cortical reaction → amphimixis → first cell division

177. A giant rat is formed in the laboratory, what is the reason :-

- (1) Gene mutation
- (2) Gene synthesis
- (3) Gene manipulation
- (4) Gene replication

178. The epithelial cells lining the stomach of vertebrate is protected from damage by HCl because :-

- (1) HCl is neutralized by gastric juice
- (2) HCl is too dilute
- (3) The epithelial cells are resistant to the action of HCl
- (4) The epithelial cells are covered by mucous secretion

179. Zona fasciculata hormones affects :-

- (1) Decrease glucose level
- (2) Increase Ca^{+2} level in blood
- (3) Parturition
- (4) RBC formation

180. *Pisum sativum* is a member of family fabaceae, which of the following character is not correct with respect to this plant :-

- (1) Alternate phyllotaxy and reticulate venation
- (2) Zygomorphic symmetry and superior ovary
- (3) Marginal placenta and valvate calyx
- (4) Monocot plant with endospermous seed



SPACE FOR ROUGH WORK

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1. Each candidate must show on demand his/her Allen ID Card to the Invigilator.
2. No candidate, without special permission of the Invigilator, would leave his/her seat.
3. The candidates should not leave the Examination Hall without handing over their Answer Sheet to the Invigilator on duty.
4. Use of Electronic/Manual Calculator is prohibited.
5. The candidates are governed by all Rules and Regulations of the examination with regard to their conduct in the Examination Hall. All cases of unfair means will be dealt with as per Rules and Regulations of this examination.
6. No part of the Test Booklet and Answer Sheet shall be detached under any circumstances.
7. The candidates will write the Correct Name and Form No. in the Test Booklet/Answer Sheet.

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